CAR BLACK BOX PROJECT

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Abstract

A Car Black Box is an event recording system used to monitor and store important vehicle events. This project is developed using the PIC16F877A microcontroller. The system records events such as Engine status, speed changes, and other vehicle-related information. The recorded data is stored in EEPROM and can be displayed on a CLCD. The stored logs can also be transmitted through UART for analysis.

Introduction

A black box is a device used to record events occurring in a vehicle. In case of accidents or failures, the recorded information can be used for investigation. This project simulates a vehicle black box system using embedded hardware and software components.

Objectives

- Record vehicle events.
- Store event logs in EEPROM.
- Display information on CLCD.
- Provide menu-based log viewing.
- Download stored logs using UART communication.
- Maintain date and time using RTC.

Components Used

Hardware

- PIC16F877A Microcontroller
- CLCD Display
- DS1307 RTC Module
- EEPROM
- Matrix Keypad
- UART Interface
- Power Supply

Software

- MPLAB X IDE

- XC8 Compiler
- PICSimLab

Working Principle

1. The system continuously monitors user inputs.
2. Event information is recorded along with date and time.
3. Logs are stored in EEPROM memory.
4. Users can view stored logs through CLCD.
5. Logs can be downloaded using UART communication.
6. RTC provides accurate date and time information.

Features

- Event recording and storage.
- Real-time clock support.
- EEPROM data storage.
- UART log download.
- User-friendly menu system.
- CLCD display interface.

Advantages

- Reliable event recording.
- Easy data retrieval.
- Low power consumption.
- Compact design.
- Cost-effective implementation.

Applications

- Automobile monitoring systems.
- Vehicle accident analysis.
- Driver behaviour monitoring.

Result

The Car Black Box system was successfully implemented using PIC16F877A. Events were recorded and stored in EEPROM with corresponding timestamps. The logs were displayed on CLCD and successfully transmitted through UART.

Conclusion

The project demonstrates the implementation of a basic vehicle event recording system. It effectively stores and retrieves event data using EEPROM and provides real-time monitoring through CLCD and UART communication. This system can be extended for real-world automotive applications.

Future Scope

- Cloud data storage.
- Accident detection sensors.

References

1. PIC16F877A Datasheet
2. DS1307 RTC Datasheet
3. EEPROM Datasheet
4. MPLAB X IDE Documentation
5. Embedded Systems Programming Manuals